

Section 9

Systems Programming ECE3502

Control Sections

- Each such control section can be loaded and relocated independently of the others.
- Different control sections are often used for subroutines or other logical subdivisions of a program.
- assembler directive: **CSECT**
 - section name CSECT
 - CSECT: separate location counter for each control section
- Symbols that are defined in one control section **cannot be used directly by another control section.**
- They must be identified as **external references** for the loader to handle.

Two assembler directives are used:

EXTDEF (external definition)

- Identify those symbols that are defined in this control section and can be used in other control sections.
- Control section names are automatically considered as external symbols.

EXTREF (external reference)

- Identify those symbols that are used in this control section but defined in other control sections.

Loc.	Symbols	Instructions	reference	Obj. code
0000	COPY	START	0	
-----		EXTDEF	BUFFER, BUFEND, LENGTH	
-----		EXTREF	RDREC, WRREC	
0000	FIRST	STL	RETADR	172027
0003	CLOOP	+JSUB	RDREC	4B100000
0007		LDA	LENGTH	032023
000A		COMP	#0	290000
000D		JEQ	ENDFIL	332007
0010		+JSUB	WRREC	4B100000
0014		J	CLOOP	3F2FEC
0017	ENDFIL	LDA	=C' EOF'	032016
001A		STA	BUFFER	0F2016
001D		LDA	#3	010003
0020		STA	LENGTH	0F200A
0023		+JSUB	WRREC	4B100000
0027		J	@RETADR	3E2000
002A	RETADR	RESW	1	-----
002D	LENGTH	RESW	1	-----
-----		LTORG		-----
0030	*	=C' EOF'		454F46
0033	BUFFER	RESB	4096	-----
1033	BUFEND	EQU	*	-----
1000	MAXLEN	EQU	BUFEND-BUFFER	-----

+JSUB RDREC									
Opcode (6)	n	i	x	b	p	e	Addr(20)		
48	1	1	0	0	0	1	00000		
0100 10	11		0001				0000 0000 0000 0000 0000		
4B100000									

+JSUB WRREC									
Opcode (6)	n	i	x	b	p	e	Addr(20)		
48	1	1	0	0	0	1	00000		
0100 10	11		0001				0000 0000 0000 0000 0000		
4B100000									

H.D.R.T.M.E Record

H.COPYXX.000000.001033
Define Record: Name of ext. symbol defined(6). Address(6)
[D. BUFFER. 000033. BUFEND. 001033. LENGTH. 00002D](#)
Refer Record: Name of ext. symbol referred (6)
[R.RDREC. WRREC](#)
T.00000.1D.1720270F2016
T.00001D.0D.010003.0F200A.4B100000.3E3000
T.000030.03.454F46
M.000004.05.+RDREC
M.000011.05.+WRREC
M.000024.05.+WRREC
E.00000

Loc.	Symbols	Instructions	reference	Obj. code
0000	RDREC	CSECT		
-----		EXTREF	BUFFER, LENGTH, BUFEND	
0000		CLEAR	X	B410
0002		CLEAR	A	B400
0004		CLEAR	S	B440
0006		LDT	MAXLEN	77201F
0009	RLOOP	TD	INPUT	E3201B
000C		JEQ	RLOOP	332FFA
000F		RD	INPUT	DB2015
0012		COMPR	A, S	A004
0014		JEQ	EXIT	332009
0017		+STCH	BUFFER, X	57900000
001B		TIXR	T	B850
001D		JLT	RLOOP	3B2FE9
0020	EXIT	+STX	LENGTH	13100000
0024		RSUB		4F0000
0027	INPUT	BYTE	X'F1'	F1
0028	MAXLEN	WORD	BUFEND-BUFFER	00000

H.D.R.T.M.E Record

H.RDRECX.000000.00002B

Refer Record: Name of ext. symbol referred (6)
R.BUFFER. LENGTH, BUFEND

T.00000.1D.B410B850
T.00001D.0E.3B2FE9.13100000.4F0000.F1.000000

M.000018.05.+ BUFFER

M.000021.05.+ LENGTH

M.000028.06.+ BUFEND

M.000028.06.- BUFFER

E.00000

Incase:
MAXLEN WORD BUFEND – RLOOP (FFFFF7)

M.000028.06.+ BUFEND

M.000028.06.- RDREC

Loc.	Symbols	Instructions	reference	Obj. code
0000	WRREC	CSECT		
-----		EXTREF	LENGTH, BUFFER	
0000		CLEAR	X	B410
0002		+LDT	LENGTH	77100000
0006	WLOOP	TD	=X'05'	E32012
0009		JEQ	WLOOP	332FFA
000C		+LDCH	BUFFER, X	53900000
0010		WD	=X'05'	DF2008
0013		TI XR	T	B850
0015		JLT	WLOOP	3B2FEE
0018		RSUB		4F0000
		END	FIRST	
001B	*	=X'05'		05

H.D.R.T.M.E Record

H.WRRECX.000000.00002B

Refer Record: Name of ext. symbol referred (6)
R. LENGTH, BUFFER

T.00000.1C.B41005

M.000003.05.+ LENGTH

M.00000D.05.+ BUFFER

E.00000